

Improving Standing Balance with Mid-Air Haptic Feedback

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Abstract

Improving standing balance to prevent falls is critical to maintaining well-being. In this paper, we propose an innovative use of mid-air haptic feedback to improve standing balance using the light touch effect, a phenomenon previously studied in the medical field. The light touch effect involves an improvement in standing balance when a wall or handrail is touched with a force of 1 N or less. To exploit this phenomenon, we use focused ultrasound to create an invisible touchable point in the air. When users touch the focused point with their hands, they experience the light touch effect that improves their balance. We conducted a user study and confirmed that the invisible light touch improved standing balance.

CCS Concepts

• Human-centered computing → Haptic devices.

Keywords

Mid-air haptics, Ultrasound-focusing device, Light touch effect, Standing balance

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1 Introduction

Fall prevention is critical to maintaining the well-being of an aging population. Improving balance and postural control through technologies that support motor control is important to ensure stability and prevent falls. To explore new technologies for improving balance, we aim to develop a novel balance enhancement system that uses ultrasound-focusing devices to control the environment and enhance the balance capabilities of users interacting with it. An ultrasound-focusing device is a technology that uses focused ultrasound to create contactless haptic feedback in mid-air [Carter

et al. 2013; Hoshi et al. 2010], allowing users to interact with virtual objects and interfaces without the need for physical contact or wearable devices.

To improve balance with ultrasound-focusing devices, we utilize the light touch effect, a phenomenon previously studied in the medical field [Jeka 1997; Jeka and Lackner 1994; Lackner et al. 2001]. Light touch refers to the phenomenon where a standing person lightly touches a stable external object, such as walls and handrails, with a finger, palm, or other body part, resulting in a reduction in postural sway. As a representative example, Jeka [Jeka and Lackner 1994] reveals that both light and forceful fingertip contact significantly reduce body sway in a closed-eye stance, indicating that tactile feedback aids in anticipatory muscle activation for maintaining balance. This effect can be induced by two factors: the presence of a stable reference point and the modulation of contact forces, resulting in tissue deformation. Even when the contact force is minimal, less than 1 N, and insufficient for mechanical support, light touch has been shown to significantly reduce postural sway. Stabilization can be achieved by stimulating a small number of fingertip receptors, and contact with a stationary spatial reference is effective. Light touch can provide anticipatory information that enables muscle innervation to reduce postural sway. In the context of balance training, it is suggested that relying on handrails or canes for full body weight is less effective than using light touch to engage lower limb muscles for balance [Oshita and Yano 2016].

In this paper, we propose Invisible Light Touch (ILT), which uses mid-air haptic feedback by focused ultrasound to generate the light touch effect, resulting in improved balance. When users touch the focal point of the mid-air haptic feedback with their hands, the light touch effect is produced because the focal point satisfies two factors for light touch: the presence of a stable reference point and the less than 1 N force feedback [Hoshi et al. 2010]. The light touch effect reduces the user's postural sway and improves their standing balance as if they were touching an invisible handrail.

Our contribution is to propose Invisible Light Touch and explore a novel support system to improve the standing balance of users with ultrasound-focusing devices. The advantages of ILT are as follows. (1) Users do not need to wear any devices on their bodies and there is no fear of infection when using in public spaces due to the non-contact haptic feedback. (2) Since the focal point of the ultrasound can be freely adjusted in 3D space, the light touch effect can be applied to various standing postures and hand positions. (3) One of the characteristics of ILT is its inability to physically support the body, creating a constant light touch scenario. This

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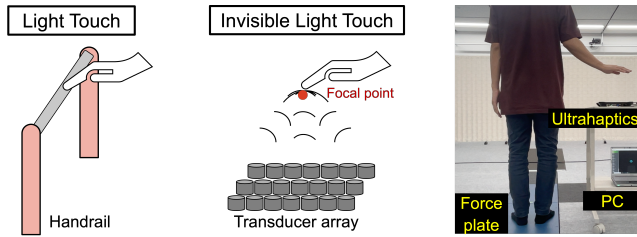


Figure 1: Comparison between light touch and Invisible Light Touch. The picture shows the experimental setup.

effectively maintains an ideal condition for balance training, which is a significant advantage for its application. These benefits indicate that the light touch effect can be manifested in a wider range of situations, suggesting that it could help improve balance in a wide range of people.

2 Invisible Light Touch

Hardware. For mid-air haptic feedback, we use an Ultrahaptics STRATOS Explore development kit (Ultraleap) [Carter et al. 2013]. Ultrahaptics consists of 256 ultrasound transducers operating at 40 kHz with an effective ultrasound focusing region $40 \times 40 \times 60 \text{ cm}^3$ above the device. The ultrasound transducer array can focus ultrasound at specific locations in 3D space and create a focal point. The Leap Motion Controller is attached to the top of the transducer array and can be used to place the focal point on a specific part of the user's hand, such as the center of the palm or the fingertips. The haptic feedback can be designed using the dedicated software, Sensation Editor.

Principle. The conditions for generating the light touch effect include touching a wall or handrail with the hand and maintaining a contact force of less than 1 N [Jeka 1997; Jeka and Lackner 1994]. In this way, slight changes in skin sensation caused by postural sway can be perceived by the high-density tactile receptors distributed in the hand, and the standing balance can be improved. To achieve this condition with an ultrasound-focusing device, the ultrasound is focused on a single point in mid-air (Figure 1). When the user touches this focal point with their hand, weak tactile feedback from the focal point can be perceived through the tactile receptors of the skin. The force generated by the focused ultrasound is less than 1 N [Hoshi et al. 2010], which is consistent with the contact force required to generate the light touch effect. When the contact situation between the focal point and the hand changes due to postural sway, the position of the focal point relative to the hand changes, and the tactile receptors in the skin can detect this spatial change, allowing the user to perceive their own postural sway more quickly than without mid-air haptic feedback, and thereby recover their balance. Since mid-air haptic feedback can satisfy the force and space conditions for the light touch effect, it can be used to improve standing balance.

3 Evaluation

We conducted a user study to evaluate the light touch effect with ILT. We recruited 29 participants (16 identified as female and 13

Table 1: Results for each indicator.

	Eyes closed		p-value
	Without ILT	With ILT	
COP path length (mm)	117.8±34.3	110.9±37.2	0.033
RMS (mm)	2.86±0.65	2.68±0.64	0.065
Rectangle area (mm ²)	142.8±81.7	125.3±65.3	0.032

identified as male, age range 20s–70s, $M = 43.9$ years, $SD = 14.5$). All participants were healthy and did not have any chronic illnesses. They signed an informed consent form before the start of the experiment. This user study was approved by our institution's ethical review board. We used an Ultrahaptics STRATOS Explore development kit to provide mid-air haptic feedback to the palm of the right hand; the focal point was set at a height of about 20 cm. The diameter of the focal point was about 20 mm. We also used a force plate to measure COP. The sampling rate of the force plate was 1000 Hz. We designed a control experiment comparing conditions with and without ILT. Participants stood with their feet shoulder-width apart with eyes closed. We measured COP three times for 30 seconds for each participant in each condition. The order was counterbalanced. We resampled the measured COP data at 100 Hz, applied a second-order low-pass Butterworth filter with a cutoff frequency of 10 Hz, and calculated a 1-second moving average. Based on previous studies evaluating standing balance from COP, we calculated three indicators for evaluation: COP path length, root mean square (RMS), and rectangle area. When the standing balance is stable, the three indicators are small.

Results. The mean and standard deviation for each indicator for each condition are shown in Table 1. Comparing each indicator without ILT and with ILT, all indicators decreased with ILT. We conducted the one-tailed paired t-test for the RMS, and the one-tailed Wilcoxon signed-rank test for the COP path length and rectangle area, depending on data normality, which was assessed using the Jarque-Bera test. We performed each test with Benjamini and Hochberg correction. There were significant differences for the COP path length ($Z = 1.84$, $p = 0.033$, $r = 0.34$) and rectangle area ($Z = 1.86$, $p = 0.032$, $r = 0.35$), and no significant difference for the RMS ($t(28) = 1.56$, $p = 0.065$, $d = 0.29$). These results suggested that participants could improve their standing balance with ILT.

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